

2025-2026 学年第二学期 线性代数期末复习材料

注意事项:

1. 请勿将本复习材料带入考场!
2. 内容均经过整理与校对,但仍可能存在错别字、内容遗漏或表述不准确等问题,欢迎指正。
3. 后续勘误及更新将同步至项目地址。
4. 项目地址: <https://1077015732hyw-beep.github.io/yenrik-exam-rescue/>
5. 祝大家复习顺利,考试顺利!



一、第一章

1. 计算二阶行列式 $\begin{vmatrix} 1 & -4 \\ 2 & 10 \end{vmatrix}$

解: $= 1 \times 10 - (-4) \times 2 = 18$

2. 计算二阶行列式 $\begin{vmatrix} 1 & -3 \\ 2 & 9 \end{vmatrix}$

解: $= 1 \times 9 - (-3) \times 2 = 15$

3. 计算三阶行列式 $\begin{vmatrix} 1 & 2 & 4 \\ -2 & 2 & 1 \\ -3 & 4 & -2 \end{vmatrix}$

(1) 解:按行展开

$$\begin{aligned} &= 1 \begin{vmatrix} 2 & 1 \\ 4 & -2 \end{vmatrix} - 2 \begin{vmatrix} -2 & 1 \\ -3 & -2 \end{vmatrix} + 4 \begin{vmatrix} -2 & 2 \\ -3 & 4 \end{vmatrix} \\ &= 2 \times (-2) - 1 \times 4 - 2[(-2) \times (-2) - 1 \times (-3)] + 4[(-2) \times 4 - 2 \times (-3)] \\ &= (-8) - 2 \times 7 + 4 \times (-2) \\ &= -8 - 14 - 8 \\ &= -30 \end{aligned}$$

(2) 解:化为上三角行列式 $\begin{vmatrix} 1 & 2 & 4 \\ -2 & 2 & 1 \\ -3 & 4 & -2 \end{vmatrix} \xrightarrow[r_3+3r_1]{r_2+2r_1} \begin{vmatrix} 1 & 2 & 4 \\ 0 & 6 & 9 \\ 0 & 10 & 10 \end{vmatrix} \xrightarrow{r_3-\frac{5}{3}r_2} \begin{vmatrix} 1 & 2 & 4 \\ 0 & 6 & 9 \\ 0 & 0 & -5 \end{vmatrix}$

此时为上三角行列式 $D = 1 \times 6 \times (-5) = -30$

(3) 解:对角线法则

$$\begin{aligned} &1 \times 2 \times (-2) + 2 \times 1 \times (-3) + 4 \times (-2) \times 4 - 4 \times 2 \times (-3) - 2 \times (-2) \times (-2) - 1 \times 1 \times 4 \\ &= -4 - 6 - 32 + 24 - 8 - 4 \\ &= -30 \end{aligned}$$

4. 计算三阶行列式 $\begin{vmatrix} 2 & 0 & 1 \\ 1 & -4 & -1 \\ -1 & 8 & -3 \end{vmatrix}$

(1) 解:按行展开

$$\begin{aligned}
 &= 2 \begin{vmatrix} -4 & -1 \\ 8 & -3 \end{vmatrix} - 0 \begin{vmatrix} 1 & -1 \\ -1 & -3 \end{vmatrix} + 1 \begin{vmatrix} 1 & -4 \\ -1 & 8 \end{vmatrix} \\
 &= 2[(-4) \times (-3) - (-1) \times 8] - 0[1 \times (-3) - (-1) \times (-1)] + 1[1 \times 8 - (-4) \times (-1)] \\
 &= 2 \times 20 + 0 + 1 \times 4 \\
 &= 40 + 4 \\
 &= 44
 \end{aligned}$$

(2) 解:化为上三角行列式

$$\begin{vmatrix} 2 & 0 & 1 \\ 1 & -4 & -1 \\ -1 & 8 & -3 \end{vmatrix} \xrightarrow[r_3 + \frac{1}{2}r_1]{r_2 - \frac{1}{2}r_1} \begin{vmatrix} 2 & 0 & 1 \\ 0 & -4 & -\frac{3}{2} \\ 0 & 8 & -\frac{5}{2} \end{vmatrix} \xrightarrow{r_3 + 2r_2} \begin{vmatrix} 2 & 0 & 1 \\ 0 & -4 & -\frac{3}{2} \\ 0 & 0 & -\frac{11}{2} \end{vmatrix}$$

此时为上三角行列式 $D = 2 \times (-4) \times (-\frac{11}{2}) = 44$

(3) 解:对角线法则

$$\begin{aligned}
 &2 \times (-4) \times (-3) + 0 \times (-1) \times (-1) + 1 \times 1 \times 8 - 1 \times (-4) \times (-1) - 0 \times 1 \times (-3) - 2 \times (-1) \times 8 \\
 &= 24 + 0 + 8 - 4 - 0 + 16 \\
 &= 44
 \end{aligned}$$

5. 在行列式中 $\begin{vmatrix} 5 & 1 & 2 \\ -4 & 2 & 3 \\ 2 & 0 & 5 \end{vmatrix}$ 元素 a_{33} 的余子式 M_{33}

解: $\begin{vmatrix} 5 & 1 & 2 \\ -4 & 2 & 3 \\ 2 & 0 & 5 \end{vmatrix}$, $M_{33} = \begin{vmatrix} 5 & 1 \\ -4 & 2 \end{vmatrix} = 5 \times 2 - 1 \times (-4) = 10 + 4 = 14$

6. 在行列式中 $\begin{vmatrix} 5 & 1 & 2 \\ -4 & 2 & 3 \\ 2 & 0 & 5 \end{vmatrix}$ 元素 a_{13} 的余子式 M_{13}

解: $\begin{vmatrix} 5 & 1 & 2 \\ -4 & 2 & 3 \\ 2 & 0 & 5 \end{vmatrix}$, $M_{13} = \begin{vmatrix} -4 & 2 \\ 2 & 0 \end{vmatrix} = (-4) \times 0 - 2 \times 2 = -4$

7. 求 23415 的逆序数

解: $\left. \begin{array}{l} 2 \text{ 前面有 } 0 \text{ 个比它大的数;} \\ 3 \text{ 前面有 } 0 \text{ 个比它大的数;} \\ 4 \text{ 前面有 } 0 \text{ 个比它大的数;} \\ 1 \text{ 前面有 } 3 \text{ 个比它大的数;} \\ 5 \text{ 前面有 } 0 \text{ 个比它大的数.} \end{array} \right\}, \therefore 0 + 0 + 0 + 3 + 0 = 3$

8. 求 43251 的逆序数

解: $\left. \begin{array}{l} 4 \text{ 前面有 } 0 \text{ 个比它大的数;} \\ 3 \text{ 前面有 } 1 \text{ 个比它大的数;} \\ 2 \text{ 前面有 } 2 \text{ 个比它大的数;} \\ 5 \text{ 前面有 } 0 \text{ 个比它大的数;} \\ 1 \text{ 前面有 } 4 \text{ 个比它大的数.} \end{array} \right\}, \therefore 0 + 1 + 2 + 0 + 4 = 7$

9. 若三阶行列式的第三行元素是 3, 1, 2 它们的代数余子式分别为 1, -3, -5 求 D

解: 根据行列式按第三行展开:

$$\begin{aligned}
 D &= 3 \times 1 + 1 \times (-3) + 2 \times (-5) \\
 &= 3 - 3 - 10
 \end{aligned}$$

$$=-10$$

10. 若三阶行列式的第三行元素是 3, 1, 2 它们的代数余子式分别为 1, 17, -5 求 D

解: 根据行列式按第三行展开:

$$D = 3 \times 1 + 1 \times 17 + 2 \times (-5)$$

$$= 3 + 17 - 10$$

$$= 10$$

11. 如果 $\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = 20$, 则 $\begin{vmatrix} 2a_{11} & 2a_{12} & 2a_{13} \\ a_{21} & a_{22} & a_{23} \\ 4a_{31} & 4a_{32} & 4a_{33} \end{vmatrix}$

$$\text{解: } 2 \times 4 \times 20 = 160$$

12. 如果 $\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = 30$, 则 $\begin{vmatrix} 2a_{11} & 2a_{12} & 2a_{13} \\ a_{21} & a_{22} & a_{23} \\ 3a_{31} & 3a_{32} & 3a_{33} \end{vmatrix}$

$$\text{解: } 2 \times 3 \times 30 = 180$$

13. 求 $\begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 10 \\ 1 & 3 & 9 & 27 \\ 1 & 4 & 16 & 70 \end{vmatrix}$

$$\text{解: } \begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 10 \\ 1 & 3 & 9 & 27 \\ 1 & 4 & 16 & 70 \end{vmatrix} \xrightarrow[r_4-r_1]{\begin{matrix} r_2-r_1 \\ r_3-r_1 \end{matrix}} \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & 9 \\ 0 & 2 & 8 & 26 \\ 0 & 3 & 15 & 69 \end{vmatrix} \xrightarrow[r_4-3r_3]{r_4-r_2} \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & 9 \\ 0 & 0 & 2 & 8 \\ 0 & 0 & 0 & 18 \end{vmatrix}$$

此时为上三角行列式,

$$D = 1 \times 1 \times 2 \times 18 = 36$$

14. 求 $\begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 5 \\ 1 & 3 & 6 & 10 \\ 1 & 4 & 12 & 23 \end{vmatrix}$

$$\text{解: } \begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 5 \\ 1 & 3 & 6 & 10 \\ 1 & 4 & 12 & 23 \end{vmatrix} \xrightarrow[r_4-r_1]{\begin{matrix} r_2-r_1 \\ r_3-r_1 \end{matrix}} \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 4 \\ 0 & 2 & 5 & 9 \\ 0 & 3 & 11 & 22 \end{vmatrix} \xrightarrow[r_4-3r_2]{r_3-2r_2} \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 5 & 10 \end{vmatrix} \xrightarrow[r_4-5r_3]{r_4-5r_2} \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 5 \end{vmatrix}$$

此时为上三角行列式,

$$D = 1 \times 1 \times 1 \times 5 = 5$$

二、第二章

15. 设 $A = \begin{pmatrix} 1 & 2 \\ 4 & 9 \end{pmatrix}$, 则 A^{-1}

解: $\because A^{-1} = \frac{1}{|A|} A^*$ (主对角线上的元素互换, 副对角线上的元素变为相反数。)

$$\therefore A^{-1} = \frac{1}{1 \times 9 - 2 \times 4} \begin{pmatrix} 9 & -2 \\ -4 & 1 \end{pmatrix} = \frac{1}{1} \begin{pmatrix} 9 & -2 \\ -4 & 1 \end{pmatrix} = \begin{pmatrix} 9 & -2 \\ -4 & 1 \end{pmatrix}$$

16. 设 $A = \begin{pmatrix} 1 & 2 \\ 6 & 13 \end{pmatrix}$, 则 A^{-1}

解: $\because A^{-1} = \frac{1}{|A|} A^*$ (主对角线上的元素互换, 副对角线上的元素变为相反数。)

$$\therefore A^{-1} = \frac{1}{1 \times 13 - 2 \times 6} \begin{pmatrix} 13 & -2 \\ -6 & 1 \end{pmatrix} = \frac{1}{1} \begin{pmatrix} 13 & -2 \\ -6 & 1 \end{pmatrix} = \begin{pmatrix} 13 & -2 \\ -6 & 1 \end{pmatrix}$$

17. 设 A 为二阶方阵, 且 $|A| = 2$, 求 $|3A^{-1}|$

解: $\because A$ 为二阶方阵,

$$\therefore |kA| = k^2 |A|$$

$$\text{且 } |A^{-1}| = \frac{1}{|A|}$$

$$\therefore |3A^{-1}| = 3^2 |A^{-1}| = 9 \times \frac{1}{|A|} = 9 \times \frac{1}{2} = \frac{9}{2}$$

18. 设 A 为二阶方阵, 且 $|A| = 1$, 求 $|3A^{-1}|$

解: $\because A$ 为二阶方阵,

$$\therefore |kA| = k^2 |A|$$

$$\text{且 } |A^{-1}| = \frac{1}{|A|}$$

$$\therefore |3A^{-1}| = 3^2 |A^{-1}| = 9 \times \frac{1}{|A|} = 9 \times \frac{1}{1} = 9$$

19. 设 $A = \begin{pmatrix} 1 & -1 & 1 \\ 2 & 0 & 1 \\ 3 & 1 & -2 \end{pmatrix}, B = \begin{pmatrix} 1 & 2 & 3 \\ 4 & -2 & 4 \\ 0 & 5 & 1 \end{pmatrix}$, 求 $3B, A - B, B^T$

$$\text{解: } 3B = 3 \begin{pmatrix} 1 & 2 & 3 \\ 4 & -2 & 4 \\ 0 & 5 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 6 & 9 \\ 12 & -6 & 12 \\ 0 & 15 & 3 \end{pmatrix}$$

$$A - B = \begin{pmatrix} 1 & -1 & 1 \\ 2 & 0 & 1 \\ 3 & 1 & -2 \end{pmatrix} - \begin{pmatrix} 1 & 2 & 3 \\ 4 & -2 & 4 \\ 0 & 5 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -3 & -2 \\ -2 & 2 & -3 \\ 3 & -4 & -3 \end{pmatrix}$$

$$B^T = \begin{pmatrix} 1 & 4 & 0 \\ 2 & -2 & 5 \\ 3 & 4 & 1 \end{pmatrix} \text{ (互换行列)}$$

20. 设 $A = \begin{pmatrix} 1 & -1 & 1 \\ 2 & 0 & 1 \\ 3 & 1 & -2 \end{pmatrix}, B = \begin{pmatrix} 1 & 2 & 3 \\ 4 & -2 & 4 \\ 0 & 5 & 1 \end{pmatrix}$, 求 $5A, A + B, A^T$

$$\text{解: } 5A = 5 \begin{pmatrix} 1 & -1 & 1 \\ 2 & 0 & 1 \\ 3 & 1 & -2 \end{pmatrix} = \begin{pmatrix} 5 & -5 & 5 \\ 10 & 0 & 5 \\ 15 & 5 & -10 \end{pmatrix}$$

$$A + B = \begin{pmatrix} 1 & -1 & 1 \\ 2 & 0 & 1 \\ 3 & 1 & -2 \end{pmatrix} + \begin{pmatrix} 1 & 2 & 3 \\ 4 & -2 & 4 \\ 0 & 5 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 4 \\ 6 & -2 & 5 \\ 3 & 6 & -1 \end{pmatrix}$$

$$A^T = \begin{pmatrix} 1 & 2 & 3 \\ -1 & 0 & 1 \\ 1 & 1 & -2 \end{pmatrix} \text{ (互换行列)}$$

21. 设 $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, 求 A^4, A^{10}, A^{11}

解:找规律: $A^2 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, A^3 = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}, A^4 = \begin{pmatrix} 1 & 4 \\ 0 & 1 \end{pmatrix}$

故 $A^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$

$\therefore A^4 = \begin{pmatrix} 1 & 4 \\ 0 & 1 \end{pmatrix}, A^{10} = \begin{pmatrix} 1 & 10 \\ 0 & 1 \end{pmatrix}, A^{11} = \begin{pmatrix} 1 & 11 \\ 0 & 1 \end{pmatrix}$

22. 设 $A = \begin{pmatrix} 1 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & 2 & 1 \end{pmatrix}$, 求 A^{-1}

解: $(A | E) = \left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 2 & 1 & 0 & 0 & 1 \end{array} \right) \xrightarrow{r_2+r_1} \left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 2 & 1 & 0 & 0 & 1 \end{array} \right) \xrightarrow{r_3-2r_2} \left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & -1 & -2 & -2 & 1 \end{array} \right)$
 $\xrightarrow{r_3 \times (-1)} \left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 2 & 2 & -1 \end{array} \right) \xrightarrow{r_2-r_3} \left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & -1 & 1 \\ 0 & 0 & 1 & 2 & 2 & -1 \end{array} \right) \xrightarrow{r_1-r_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & 1 & -1 \\ 0 & 1 & 0 & -1 & -1 & 1 \\ 0 & 0 & 1 & 2 & 2 & -1 \end{array} \right) = (E | A^{-1})$
 $\therefore A^{-1} = \begin{pmatrix} 2 & 1 & -1 \\ -1 & -1 & 1 \\ 2 & 2 & -1 \end{pmatrix}$

三、第三章

23. 判定向量组 $\alpha_1, \alpha_2, \alpha_3$ 是线性相关还是线性无关 $\alpha_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \alpha_2 = \begin{pmatrix} 3 \\ 5 \\ 4 \end{pmatrix}, \alpha_3 = \begin{pmatrix} 5 \\ -1 \\ 6 \end{pmatrix}$

解: $A = (\alpha_1 \ \alpha_2 \ \alpha_3) = \begin{pmatrix} 1 & 3 & 5 \\ 1 & 5 & -1 \\ 1 & 4 & 6 \end{pmatrix} \xrightarrow{r_2-r_1} \begin{pmatrix} 1 & 3 & 5 \\ 0 & 2 & -6 \\ 0 & 1 & 1 \end{pmatrix} \xrightarrow{r_3-r_1} \begin{pmatrix} 1 & 3 & 5 \\ 0 & 2 & -6 \\ 0 & 1 & 1 \end{pmatrix} \xrightarrow{r_3 \leftrightarrow r_2} \begin{pmatrix} 1 & 3 & 5 \\ 0 & 1 & 1 \\ 0 & 2 & -6 \end{pmatrix} \xrightarrow{r_3-2r_2} \begin{pmatrix} 1 & 3 & 5 \\ 0 & 1 & 1 \\ 0 & 0 & -8 \end{pmatrix}$

$\therefore R(A) = 3$ 与向量组向量的个数相等

$\therefore \alpha_1, \alpha_2, \alpha_3$ 线性无关

24. 判定向量组 $\alpha_1, \alpha_2, \alpha_3$ 是线性相关还是线性无关 $\alpha_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \alpha_2 = \begin{pmatrix} 3 \\ 4 \\ 7 \end{pmatrix}, \alpha_3 = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$

解: $A = (\alpha_1 \ \alpha_2 \ \alpha_3) = \begin{pmatrix} 1 & 3 & 2 \\ 1 & 4 & -1 \\ 1 & 7 & 3 \end{pmatrix} \xrightarrow{r_2-r_1} \begin{pmatrix} 1 & 3 & 2 \\ 0 & 1 & -3 \\ 0 & 4 & 1 \end{pmatrix} \xrightarrow{r_3-r_1} \begin{pmatrix} 1 & 3 & 2 \\ 0 & 1 & -3 \\ 0 & 4 & 1 \end{pmatrix} \xrightarrow{r_3-4r_2} \begin{pmatrix} 1 & 3 & 2 \\ 0 & 1 & -3 \\ 0 & 0 & 13 \end{pmatrix}$

$\therefore R(A) = 3$ 与向量组向量的个数相等

$\therefore \alpha_1, \alpha_2, \alpha_3$ 线性无关

25. 设 A 为 n 阶方阵, 证明: $A^2 - 3A - 10E = O$, 则 $R(A+2E) + R(A-5E) = n$

证明: $A^2 - 3A - 10E = O$

因式分解得 $(A+2E)(A-5E) = O$

若 $A_{m \times n} B_{n \times l} = O$, 则 $R(A) + R(B) \leq n$

$\therefore R(A+2E) + R(A-5E) \leq n$

又 $\therefore R(A+B) \leq R(A) + R(B)$

$\therefore R(A+2E) + R(A-5E) = R(A+2E) + R(5E-A) \geq R[(A+2E) + (5E-A)] = R(7E) = n$

综上: $R(A+2E) + R(A-5E) = n$

26. 设 A 为 n 阶方阵, 证明: $A^2 - 3A - 4E = O$, 则 $R(A+E) + R(A-4E) = n$

证明: $A^2 - 3A - 4E = O$

因式分解得 $(A+E)(A-4E) = O$

若 $A_{m \times n} B_{n \times l} = O$, 则 $R(A) + R(B) \leq n$

$\therefore R(A+E) + R(A-4E) \leq n$

又 $\because R(A+B) \leq R(A) + R(B)$

$\therefore R(A+E) + R(A-4E) = R(A+E) + R(4E-A) \geq R[(A+E) + (4E-A)] = R(5E) = n$

综上: $R(A+E) + R(A-4E) = n$

27. λ 取何值时, 非齐次线性方程组 $\begin{pmatrix} 1 & \lambda-1 & -2 \\ 0 & \lambda-2 & \lambda+1 \\ 0 & 0 & 2\lambda+1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 5 \end{pmatrix}$, (1)有惟一解, (2)无解, (3)有无限

多解? 并在无限多解时求出通解

解: $|A| = \begin{vmatrix} 1 & \lambda-1 & -2 \\ 0 & \lambda-2 & \lambda+1 \\ 0 & 0 & 2\lambda+1 \end{vmatrix} = (\lambda-2)(2\lambda+1)$

(1) 当 $|A| \neq 0$, 即 $\lambda \neq 2$ 且 $\lambda \neq -\frac{1}{2}$ 时。方程组有惟一解

(2) 当 $\lambda = -\frac{1}{2}$ 时。增广矩阵 $B = \begin{pmatrix} 1 & -\frac{3}{2} & -2 & 1 \\ 0 & -\frac{5}{2} & \frac{1}{2} & 3 \\ 0 & 0 & 0 & 5 \end{pmatrix}$,

$\therefore R(A) = 2, R(B) = 3, R(A) \neq R(B)$, 方程组无解

(3) 当 $\lambda = 2$ 时, 增广矩阵 $B = \begin{pmatrix} 1 & 1 & -2 & 1 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 5 & 5 \end{pmatrix} \xrightarrow{5r_2-3r_3} \begin{pmatrix} 1 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 5 & 5 \end{pmatrix} \xrightarrow{r_2 \leftrightarrow r_3} \begin{pmatrix} 1 & 1 & -2 & 1 \\ 0 & 0 & 5 & 5 \\ 0 & 0 & 0 & 0 \end{pmatrix}$

$\xrightarrow{r_2 \times \frac{1}{5}} \begin{pmatrix} 1 & 1 & -2 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{r_1+2r_2} \begin{pmatrix} 1 & 1 & 0 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$, $\therefore R(A) = R(B) = 2 < 3$, 有无限多解

即 $\begin{cases} x_1 + x_2 = 3 \\ x_3 = 1 \end{cases}$, 设自由未知量 $x_2 = c$

$\therefore$ 通解为 $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -c+3 \\ c \\ 1 \end{pmatrix} = c \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix}, (c \in \mathbb{R})$

四、第四章

28. 向量组 A 能由向量组 B 线性表示的充要条件 $R(B) = R(B, A)$

29. 向量组 B 能由向量组 A 线性表示的充要条件 $R(A) = R(A, B)$

30. 求解非齐次线性方程组 $\begin{cases} x_1 - 2x_2 + 3x_3 - x_4 = 1 \\ 3x_1 - 5x_2 + 5x_3 + 4x_4 = 4 \\ 2x_1 - 3x_2 + 2x_3 + 5x_4 = 3 \end{cases}$

解: $B = \begin{pmatrix} 1 & -2 & 3 & -1 & 1 \\ 3 & -5 & 5 & 4 & 4 \\ 2 & -3 & 2 & 5 & 3 \end{pmatrix} \xrightarrow{r_2-3r_1, r_3-2r_1} \begin{pmatrix} 1 & -2 & 3 & -1 & 1 \\ 0 & 1 & -4 & 7 & 1 \\ 0 & 1 & -4 & 7 & 1 \end{pmatrix} \xrightarrow{r_3-r_2} \begin{pmatrix} 1 & -2 & 3 & -1 & 1 \\ 0 & 1 & -4 & 7 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$

$$\sim \begin{pmatrix} 1 & 0 & -5 & 13 & 3 \\ 0 & 1 & -4 & 7 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \therefore R(A) = R(B) = 2 < 4, \text{有无穷多解}$$

$$\therefore \begin{cases} x_1 - 5x_3 + 13x_4 = 3 \\ x_2 - 4x_3 + 7x_4 = 1 \end{cases}, \text{设自由未知量 } x_3 = c_1, x_4 = c_2$$

$$\therefore \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 5c_1 - 13c_2 + 3 \\ 4c_1 - 7c_2 + 1 \\ c_1 \\ c_2 \end{pmatrix} = c_1 \begin{pmatrix} 5 \\ 4 \\ 1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} -13 \\ -7 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 3 \\ 1 \\ 0 \\ 0 \end{pmatrix}, (c_1, c_2 \in \mathbb{R})$$

31. 求解非齐次线性方程组
$$\begin{cases} x_1 + 4x_2 - 3x_3 + 5x_4 = -2 \\ 2x_1 + 9x_2 - x_3 + x_4 = 2 \\ 3x_1 + 14x_2 + x_3 - 3x_4 = 6 \end{cases}$$

$$\text{解: } B = \begin{pmatrix} 1 & 4 & -3 & 5 & -2 \\ 2 & 9 & -1 & 1 & 2 \\ 3 & 14 & 1 & -3 & 6 \end{pmatrix} \xrightarrow[r_3-3r_1]{r_2-2r_1} \begin{pmatrix} 1 & 4 & -3 & 5 & -2 \\ 0 & 1 & 5 & -9 & 6 \\ 0 & 2 & 10 & -18 & 12 \end{pmatrix} \xrightarrow{r_3-2r_2} \begin{pmatrix} 1 & 4 & -3 & 5 & -2 \\ 0 & 1 & 5 & -9 & 6 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow{r_1-4r_2} \begin{pmatrix} 1 & 0 & -23 & 41 & -26 \\ 0 & 1 & 5 & -9 & 6 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, R(A) = R(B) = 2 < 4$$

$$\therefore \begin{cases} x_1 - 23x_3 + 41x_4 = -26 \\ x_2 + 5x_3 - 9x_4 = 6 \end{cases}, \text{设自由未知量 } x_3 = c_1, x_4 = c_2, \text{则} \begin{cases} x_1 = 23c_1 - 41c_2 - 26 \\ x_2 = -5c_1 + 9c_2 + 6 \end{cases}$$

$$\therefore \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 23c_1 - 41c_2 - 26 \\ -5c_1 + 9c_2 + 6 \\ c_1 \\ c_2 \end{pmatrix} = c_1 \begin{pmatrix} 23 \\ -5 \\ 1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} -41 \\ 9 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} -26 \\ 6 \\ 0 \\ 0 \end{pmatrix}, (c_1, c_2 \in \mathbb{R})$$

32. 利用初等行变换求矩阵 A 的列向量组的秩, 并求一个最大无关组, 并把其余列向量用最大无关组线性表示

$$A = (\alpha_1 \quad \alpha_2 \quad \alpha_3 \quad \alpha_4) = \begin{pmatrix} 1 & 1 & 2 & 1 \\ 0 & 2 & 1 & 1 \\ 2 & 0 & 3 & 1 \\ 1 & 1 & 0 & 0 \end{pmatrix}$$

$$\text{解: } A \xrightarrow[r_4-r_1]{r_3-2r_1} \begin{pmatrix} 1 & 1 & 2 & 1 \\ 0 & 2 & 1 & 1 \\ 0 & -2 & -1 & -1 \\ 0 & 0 & -2 & -1 \end{pmatrix} \xrightarrow{r_3+r_2} \begin{pmatrix} 1 & 1 & 2 & 1 \\ 0 & 2 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -2 & -1 \end{pmatrix} \xrightarrow{r_3 \leftrightarrow r_4} \begin{pmatrix} 1 & 1 & 2 & 1 \\ 0 & 2 & 1 & 1 \\ 0 & 0 & -2 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow[-\frac{1}{2}r_3]{\frac{1}{2}r_2} \begin{pmatrix} 1 & 1 & 2 & 1 \\ 0 & 1 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow{r_2-\frac{1}{2}r_3} \begin{pmatrix} 1 & 1 & 2 & 1 \\ 0 & 1 & 0 & \frac{1}{4} \\ 0 & 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{r_1-2r_3} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{4} \\ 0 & 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{r_1-r_2} \begin{pmatrix} 1 & 0 & 0 & -\frac{1}{4} \\ 0 & 1 & 0 & \frac{1}{4} \\ 0 & 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$\therefore R(A) = 3$, 且各行首非零元分别位于第 1, 2, 3 列

故矩阵 A 的第 1, 2, 3 列向量线性无关。

$$\therefore \alpha_1, \alpha_2, \alpha_3 \text{ 为列向量组的最大无关组, } \alpha_4 = -\frac{1}{4}\alpha_1 + \frac{1}{4}\alpha_2 + \frac{1}{2}\alpha_3$$

33. 利用初等行变换求矩阵 A 的列向量组的秩, 并求一个最大无关组, 并把其余列向量用最大无关组线性表示

$$A = (\alpha_1 \quad \alpha_2 \quad \alpha_3 \quad \alpha_4) = \begin{pmatrix} 1 & 1 & 2 & 3 \\ 0 & 2 & 1 & 3 \\ 2 & 0 & 3 & 3 \\ 1 & 1 & 0 & 1 \end{pmatrix}$$

$$\text{解: } A \xrightarrow[r_4-r_1]{r_3-2r_1} \begin{pmatrix} 1 & 1 & 2 & 3 \\ 0 & 2 & 1 & 3 \\ 0 & -2 & -1 & -3 \\ 0 & 0 & -2 & -2 \end{pmatrix} \xrightarrow{r_3+r_2} \begin{pmatrix} 1 & 1 & 2 & 3 \\ 0 & 2 & 1 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -2 & -2 \end{pmatrix} \xrightarrow{r_3 \leftrightarrow r_4} \begin{pmatrix} 1 & 1 & 2 & 3 \\ 0 & 2 & 1 & 3 \\ 0 & 0 & -2 & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow[-\frac{1}{2}r_3]{\frac{1}{2}r_2} \begin{pmatrix} 1 & 1 & 2 & 3 \\ 0 & 1 & \frac{1}{2} & \frac{3}{2} \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow[r_2-\frac{1}{2}r_3]{r_1-\frac{1}{2}r_3} \begin{pmatrix} 1 & 1 & 2 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{r_1-2r_3} \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{r_1-r_2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$\therefore R(A) = 3$, 且各行首非零元分别位于第 1, 2, 3 列

故矩阵 A 的第 1, 2, 3 列向量线性无关。

$\therefore \alpha_1, \alpha_2, \alpha_3$ 为列向量组的最大无关组, $\alpha_4 = \alpha_2 + \alpha_3$